

AI Engineer Intern – Take-Home Exercise

Comparative Financial Analysis Dashboard - US Public Companies

CustomerInsights.AI, Inc.

Timeline: EOD Friday, June 5

Submit your responses to: careers@customerinsights.ai

Read this first

A working RAG chatbot paired with a clean dashboard is the baseline expectation. What truly distinguishes strong submissions is what you build beyond that foundation: the accuracy of your numbers, the rigor of your evaluation, and how your system behaves when data is incomplete, inconsistent, or missing. Build the happy path quickly using tools you're comfortable with, and focus your effort on solving the more nuanced challenges.

This exercise is intentionally broader than can be fully completed within the allotted time. That's by design. We are much more interested in how you prioritize, the tradeoffs you make, and how clearly you communicate your reasoning. A focused, well-executed subset will stand out far more than an attempt to cover everything superficially.

The task

Choose **3–5 publicly traded US companies** ideally competitors in the same sector, so comparison is meaningful (e.g., a few large-cap retailers, banks, or semiconductor firms). Gather **10–15 public filings** for them from SEC EDGAR - 10-Ks and 10-Qs spanning **multiple years**, so trends and comparisons are possible.

Build a **Streamlit dashboard** that doesn't just retrieve facts but **analyzes** them: it should extract financial figures, compute derived metrics, compare companies against each other and over time, link material changes back to management's narrative, and answer analytical questions with every number and claim traceable to its source.

Baseline table stakes

- A Streamlit app that ingests your chosen filings (any format works - PDF, HTML, or XBRL).
- Natural-language Q&A over the corpus (RAG) with cited sources.
- Comparative visualizations metrics charted across your chosen companies and over time (e.g., revenue, margins, or leverage).
- Some agentic behavior is beyond single-shot retrieval query rewriting, retrieval verification, self-checking, or tool use.

- **Use an LLM framework you're comfortable with** (LangChain, LangGraph, LlamaIndex, or similar) for orchestration and retrieval. We want to see you work inside real tooling rather than reinvent the plumbing use whichever you know best.

Where the real points are

1. Analytical depth, not just retrieval

This is where we most want to see your thinking. Your dashboard should go beyond pulling figures and reason over them the LLM can do the analytical reasoning, working from figures drawn directly from the filings. A few things we'd love to see (pick what you can do well):

- **Derived metrics:** compute things that aren't printed directly in the filings e.g., revenue growth (YoY and/or CAGR), gross/operating/net margins, leverage (debt-to-equity), or free cash flow. These must be calculated from extracted figures, with the inputs and formula reproducible and traceable to source.
- **Comparison:** benchmark your companies against each other and across time who is outperforming, on what, and by how much.
- **Quant-to-narrative linkage:** when a metric moves materially, connect it to the relevant management commentary (MD&A, risk factors) e.g., "operating margin fell ~4 pts; management attributes it to ..." Grounded, and genuinely analytical.

2. Verifiable, grounded numbers

Every figure extracted **or computed** must be traceable to a source document and page/table, and computed metrics must show their underlying inputs. The evidence lives in the filings the system's job is to retrieve, surface, and cite it accurately, not to produce it. **We'll sanity-check a few numbers and calculations against the original filings, so accuracy matters more than how many metrics you produce a smaller set you've verified is better than a large set you haven't.** Pulling clean figures out of financial filings which come as PDF, HTML, and XBRL on EDGAR — is a real part of the challenge; use whatever formats work best for you.

3. Evaluation and transparency

Create a simple evaluation framework:

- Develop your own labeled question set
- Report:
 - Answer correctness and citation accuracy
 - Hallucination rate for unanswerable questions

We value honest, transparent evaluation over perfect scores.

4. Handle the messy reality of financial data

- **Unanswerable questions** → the system must refuse, not invent a number.

- **Conflicting / restated figures** → when filings disagree (restatements, different fiscal years, segment vs. consolidated, different units), does your system silently pick one, or surface the conflict? Show what it does.

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5. Writeup (WRITEUP.md, ~1 page)

Your writeup should briefly address:

- What is the architecture approach and files you have considered
- What is your hallucination rate, and would you trust this dashboard in front of an executive? What do you fix first?
- What's the most interesting insight your dashboard surfaced about these companies and how confident are you that it's correct?
- Show one failure you found and diagnosed a wrong number, a bad retrieval, or a unit mix-up.
- How did you use AI tools on this task, and name one specific place where you overrode or corrected what the AI suggested.
- Where did your framework help you, and where did you fight it or drop down to raw API call and why?

Constraints

- Use your normal AI tooling to write the code we want to see your real workflow. You own every line and must be able to explain it.
- Any LLM / embedding provider is fine. Keep costs low; small models are encouraged.
- All documents must be public. Cite where you obtained them.
- Don't over-build. We'd rather see a small system you can reason about than a large one you can't.

Deliverables

Submit a Git repository or a zip file containing:

- The Streamlit application.
- A README: how to run it, your design choices and why, and your document sources.
- Your evaluation results (the numbers, plus a short interpretation).
- WRITEUP.md (the judgment writeup above)

Timing

Plan to spend a few focused hours on this exercise—there is no expectation to complete everything. You will have until the deadline noted in your invitation.

Prioritize what matters most, and be transparent about what you did not have time to complete.

A clear, well-reasoned analysis is far more important than visual polish.